

Distributed Flânerie: Electronic Orbits of Memory as a Spatial Soft Montage

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Figure 1: Exhibition view of *Return to the North: Virtual Archaeology of Electronic Orbits*, showing a participant in the VR experience (left). Projected reconstruction of the Erqi Tower scene in Zhengzhou, presented on-site as part of the artwork (right).

ABSTRACT

Return to the North: Virtual Archaeology of Electronic Orbits is a 6DoF XR interactive narrative set in post-industrial railway cities in northern China. It examines how rail connectivity reshaped everyday visual culture amid long-term extraction and depletion. We assemble public media footage, personal archives, printed atlases, handheld LiDAR scans, and photogrammetric captures into a traversable layered space using SfM/MVS and 3D Gaussian Splatting (3DGS) for real-time rendering. A provenance-preserving strategy keeps differences in era, resolution, and context perceptible as volumetric fault lines, ghosting, captions, and seams—forming what we call a Digital Volumetric Layering aesthetic. Through embodied drift beyond prescribed paths, viewers infer relations among fragments, tracing how economic geography and memory politics surface in the seams of reconstruction.

Index Terms: multi-user VR, collective memory, spatial montage, provenance-aware reconstruction, 3D Gaussian Splatting (3DGS), gaze interaction.

1 PROJECT URL

<https://studioaniverse.github.io/re2north>

2 DESCRIPTION OF ARTWORK

Return to the North is situated in railway cities shaped by industrial expansion, extraction, and redevelopment. Rail infrastruc-

tures reorganized everyday rhythms and shared landmarks; many ordinary civic/industrial sites accumulated monument-like weight through repeated use and iconic imagery. We refer to such locations—department stores, squares, factories, amusement parks, workers' housing—as the monumental everyday. Today they persist as partial remnants, repurposed shells, or erased foundations, yet also as dispersed “data shadows” traveling across media platforms and archives. The artwork turns these shadows into a navigable memory landscape. Viewers traverse “electronic orbits” linking reconstructed scenes; layered volumes allow multiple times and sources to co-exist without collapsing into a single authoritative past. Fragments surface and fade through movement and attention, staging remembrance as ongoing recomposition. Single-user and multi-user modes (up to four participants) support shared drifting, while region-limited audio preserves partial separation.

Our material is drawn from field visits and archival collection across northern Chinese railway cities, including Shenyang, Shijiazhuang, Xuzhou, and Zhengzhou. In this region, the historical growth of the rail network reshaped everyday visual culture—how landmarks were photographed, circulated, and remembered—by standardizing routes, viewpoints, and iconic scenes. Yet this connectivity also carried a paradox: the same infrastructure that linked towns and enabled mobility

2.1 Design Concept: Distributed Flânerie, Railway Gaze, and Spatial Soft Montage

Rather than stabilizing memory through a unified monument or a seamless replica, the project treats collective memory in rapidly transforming railway cities as discontinuous, contested, and dynamic. As infrastructures are demolished, refurbished, or overwritten, memory survives as layered residues rather than a single recoverable image. This instability produces fractured landscapes and rewritten everyday environments that the work approaches as a spatial problem of viewing and inference.

To support a layered XR environment, we read the work through soft montage [1]. Heterogeneous images remain in juxtaposition so interpretation stays relational and revisitable. In 6DoF, montage

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moves from editing to movement: viewers compose relations by drifting, lingering, revisiting, and comparing, while meaning appears at seams where layers align, interfere, or refuse convergence. We describe the visible result of this strategy as a Digital Volumetric Layering aesthetic: multi-source images persist as semi-transparent volumetric strata where seams, density differences, and source marks remain legible—neither collapsing into a single past nor being cleaned away as noise.

The approach also inverts a historically conditioned railway gaze—windowed, guided, time-linear—commonly associated with modern travel’s prescribed trajectories [4]. Here the viewer is not a passenger carried along a directed view, but a drifter who can deviate, pause, and recompose across temporalities. Off-path looking becomes “thinking with images,” inviting viewers to trace how the rail network’s economic geography and memory politics surface through the montage of fragments.

2.2 Technical Implementation: Source-Preserving Reconstruction as Layered Space

Recent work demonstrates Gaussian Splatting workflows for immersive storytelling [2]. We adopt this direction, prioritizing provenance legibility.

2.2.1 Data sources and provenance-aware orchestration

The pipeline aggregates public media, personal archives, printed imagery, and site captures (handheld LiDAR, targeted photogrammetry). Source footage is extracted from its original spatial/temporal context to build 3D scenes yet keeps provenance cues: family portraits retain figures, public-news stills preserve floating text, and city-promotional slogans remain embedded as part of overlapping textures. Together these form the Digital Volumetric Layering aesthetic while also driving layer management and interaction triggers.

2.2.2 SfM/MVS + 3DGS

Large video sets undergo motion/blur-aware keyframe extraction. SfM/MVS recovers calibrated views and sparse structure; scenes are rendered with 3D Gaussian Splatting [3] to keep anisotropic splats interactive. Because monuments change and viewpoints span decades, reconstructions intentionally preserve mismatches. For example, combining a factory gate front sourced from a 1975 group portrait with a side view from a propaganda film can yield stretched alignments or doubled edges. Pre-digital archives also cluster around ceremonial angles (gateways, statue fronts), producing higher view-density there than in under-documented areas. These “oddities” are treated as visible traces of circulation rather than errors. Reconstruction and splat optimization are performed offline; at runtime we render the resulting 3DGS representation at interactive rates.

2.2.3 LiDAR point clouds and photogrammetric PBR anchors

Handheld LiDAR captures surviving interiors and industrial relics at architectural scale; point clouds may remain visible as “decay layers” whose grain and holes mark incomplete capture. Photogrammetry provides lightweight mesh anchors (with PBR textures) for material details such as cracks, stains, and worn signage.

2.2.4 Layer-aware opacity and compression balancing

To keep stacked imagery legible, we balance texture alpha, overlap count, and compression quality, maintaining an in-between state—part photographic, part sculptural—where source marks remain visible without overwhelming the view. Source metadata guides these adjustments so era-specific attributes stay perceptible.

2.2.5 Runtime rendering, interaction, and multi-user sync

Runtime centers on smooth 6DoF roaming through reconstructed nodes. We render 3D Gaussian Splatting (3DGS) scenes via GPU-accelerated Gaussian splat rendering, alongside visible LiDAR point-cloud layers where appropriate. Standard performance measures (e.g., culling and conservative level-of-detail) are used to keep VR playback stable. Interaction is lightweight and optional: gaze dwell or proximity can reveal captions, archival overlays, or localized audio. Multi-user mode (up to four) supports co-presence and shared exploration by synchronizing simple navigation and trigger states, without streaming heavy scene data.

3 CONCLUSION

Return to the North builds a provenance-preserving XR memory landscape. SfM/MVS, 3DGS, LiDAR, and photogrammetry keep source differences legible as a Digital Volumetric Layering aesthetic; meaning emerges through 6DoF drift as spatial soft montage. The contribution is an XR method that treats reconstruction artifacts as readable evidence, turning montage seams into navigational cues for interpreting infrastructural history.

ARTIST INFORMATION

Yin Bing is a Chinese animation artist and researcher whose interdisciplinary practice integrates experimental animation, scientific visualization, and immersive storytelling. At the Institute of Media and Interaction, China Academy of Art, he has developed pioneering frameworks for Distributed Embodied Narratives through his research in Expanded Animation.

Angela Chulei Tang is a multimedia artist and researcher whose practice investigates digital re-mediations of cultural memory, spatial narrative, and architectural heritage through immersive media and computational visualization.

Yuan Zhang is a visual artist and transmedia storyteller. She serves as a researcher and project leader at the Institute of Creative New Media and Performing Arts, Tsinghua University. Her research focuses on immersive new media art space experiences within the realm of cultural heritage.

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Ruiyao Zhu is a digital animation artist with expertise in integrating diverse digital tools to develop immersive experiences.

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